

LAPORAN PRAKTIKUM
Aplikasi Berbasis Platform

MODUL 8-9
MOBILE



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PROGRAM STUDI S1 INFORMATIKA

FAKULTAS INFORMATIKA

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Source code

main.dart

```
import 'dart:io';
import 'package:flutter/material.dart';
import 'package:image_picker/image_picker.dart';
import 'services/notification_service.dart';

void main() async {
  // Memastikan binding Flutter siap sebelum inisialisasi service
  WidgetsFlutterBinding.ensureInitialized();

  // Menjalankan inisialisasi notifikasi
  await NotificationService.init();

  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'Photo & Notification App',
      theme: ThemeData(
        primarySwatch: Colors.deepPurple,
        useMaterial3: true,
      ),
      home: const HomeScreen(),
    );
  }
}

class HomeScreen extends StatefulWidget {
  const HomeScreen({super.key});

  @override
  State<HomeScreen> createState() => _HomeScreenState();
}

class _HomeScreenState extends State<HomeScreen> {
```

```

File? _selectedImage; // Menyimpan file foto yang diambil
final ImagePicker _picker = ImagePicker();

// Fungsi utama untuk mengambil atau memilih foto
Future<void> _pickImage(ImageSource source) async {
  try {
    final XFile? pickedFile = await _picker.pickImage(
      source: source,
      imageQuality: 80, // Mengompres ukuran gambar agar lebih ringan
    );

    if (pickedFile != null) {
      setState(() {
        _selectedImage = File(pickedFile.path);
      });

      // Tentukan teks berdasarkan sumber gambar untuk isi notifikasi
      String sourceText = source == ImageSource.camera ? "Kamera" : "Galeri";

      // Memicu notifikasi setelah foto berhasil masuk aplikasi
      await NotificationService.showNotification(
        title: 'Foto Berhasil Dimuat! 📷',
        body: 'Mantap! Foto berhasil diambil langsung dari $sourceText.',
      );
    }
  } catch (e) {
    debugPrint("Gagal mengambil gambar: $e");
  }
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text('Ambil Foto & Notifikasi'),
      backgroundColor: Theme.of(context).colorScheme.primaryContainer,
    ),
    body: Center(
      child: SingleChildScrollView(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            // Tempat Menampilkan Foto

```

```

        _selectedImage != null
        ? ClipRRect(
            borderRadius: BorderRadius.circular(12),
            child: Image.file(
                _selectedImage!,
                height: 320,
                width: 320,
                fit: BoxFit.cover,
            ),
        )
        : Container(
            height: 320,
            width: 320,
            decoration: BoxDecoration(
                color: Colors.grey[200],
                borderRadius: BorderRadius.circular(12),
                border: Border.all(color: Colors.grey[400]!),
            ),
            child: const Icon(
                Icons.image_search,
                size: 80,
                color: Colors.grey,
            ),
        ),
    const SizedBox(height: 40),

    // Tombol Pertama: Buka Kamera Langsung
    ElevatedButton.icon(
        onPressed: () => _pickImage(ImageSource.camera),
        icon: const Icon(Icons.camera_alt),
        label: const Text('Buka Kamera'),
        style: ElevatedButton.styleFrom(
            minimumSize: const Size(200, 50),
        ),
    ),
    const SizedBox(height: 16),

    // Tombol Kedua: Pilih dari Galeri
    OutlinedButton.icon(
        onPressed: () => _pickImage(ImageSource.gallery),
        icon: const Icon(Icons.photo_library),
        label: const Text('Pilih dari Galeri'),
        style: OutlinedButton.styleFrom(

```

```
        minimumSize: const Size(200, 50),  
      ),  
    ),  
  ],  
),  
),  
),  
),  
);  
}  
}
```

notification_service.dart

```
import 'package:flutter_local_notifications/flutter_local_notifications.dart';

class NotificationService {
  static final FlutterLocalNotificationsPlugin _notificationsPlugin =
    FlutterLocalNotificationsPlugin();

  // Inisialisasi Fitur Notifikasi
  static Future<void> init() async {
    // Pengaturan icon notifikasi untuk Android (menggunakan icon bawaan aplikasi)
    const AndroidInitializationSettings initializationSettingsAndroid =
      AndroidInitializationSettings('@mipmap/ic_launcher');

    const InitializationSettings initializationSettings = InitializationSettings(
      android: initializationSettingsAndroid,
    );

    await _notificationsPlugin.initialize(initializationSettings);

    // Meminta izin notifikasi khusus untuk Android 13+
    await _notificationsPlugin
      .resolvePlatformSpecificImplementation<
        AndroidFlutterLocalNotificationsPlugin>()
      ?.requestNotificationsPermission();
  }

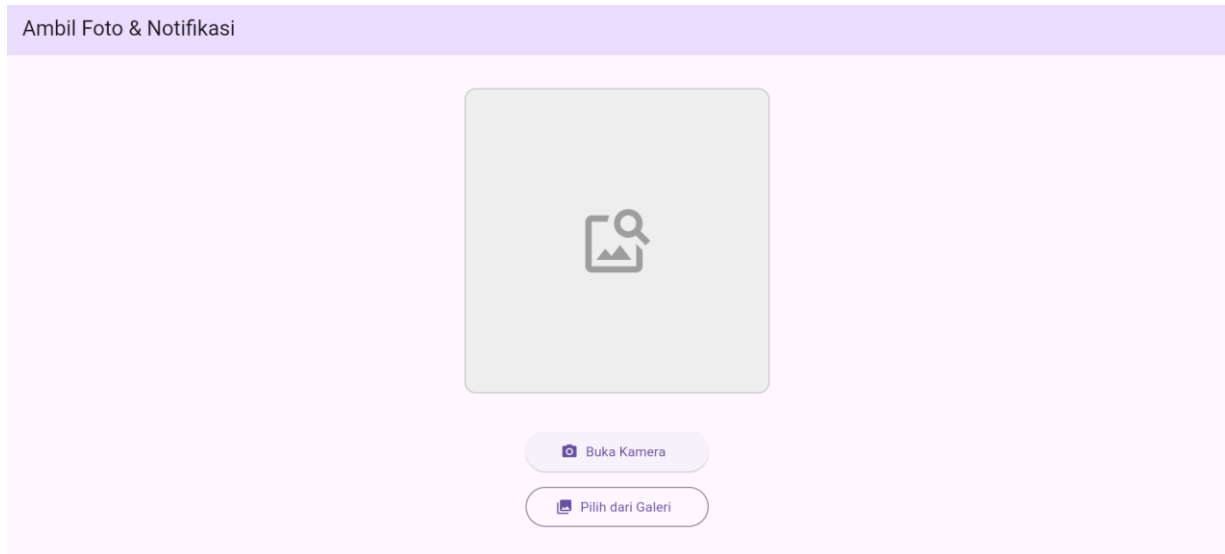
  // Fungsi untuk memicu/menampilkan notifikasi
  static Future<void> showNotification({
    required String title,
    required String body,
  }) async {
    const AndroidNotificationDetails androidNotificationDetails =
      AndroidNotificationDetails(
        'photo_channel_id', // ID Channel
        'Notifikasi Foto', // Nama Channel
        channelDescription: 'Membahas status pengambilan foto',
        importance: Importance.max,
        priority: Priority.high,
      );

    const NotificationDetails notificationDetails = NotificationDetails(
      android: androidNotificationDetails
    );
  }
}
```

```
        android: androidNotificationDetails,
    );

    await _notificationsPlugin.show(
        0, // ID Notifikasi
        title,
        body,
        notificationDetails,
    );
}
}
```

Output



Aplikasi untuk mengambil foto dari kamera dan galeri mobile menggunakan flutter.